

Training Accuracy Analysis

1. Was There an Explicit Accuracy Target?

No. The project documentation (README.md, training scripts, config files) does not define a specific accuracy target or benchmark for the model. The training was budget-constrained (\$45 total: \$15 Lightning + \$30 Modal), not accuracy-constrained.

2. What Accuracy Is Typical for LLM Fine-Tuning?

Task Type	Typical Range	Our Result
Instruction Following	75-90%	89.2%
JSON Format Compliance	85-95%	N/A (not measured)
Trading Signal Accuracy	51-60% (profitable)	89.2% (synthetic)
Domain Expertise	70-85%	89.2%

Our 89.2% accuracy is on the higher end for instruction-following fine-tuning. However, this measures token-level prediction accuracy on synthetic test data, not real-world trading profitability.

3. Why Training Accuracy Does Not Equal Trading Profit

Training accuracy measures: Given a market context and the expected JSON output, how often does the model predict the correct next token?

Trading profit requires: Given a LIVE market context with NO predetermined answer, does the model output a trade that makes money?

Key difference: Training accuracy tests pattern matching. Trading tests judgment.

Metric	What It Measures
Train Loss	How well model fits training data
Eval Loss	How well model generalizes to held-out data
Token Accuracy	Next-token prediction correctness
Trading P/L	Real-world profit (the only metric that matters)

4. What Matters More Than Training Accuracy

For a trading model, these factors matter more than the 89.2% training accuracy:

Risk Management

Does the model avoid oversized positions? Does it respect stop losses?

Regime Detection

Does the model correctly identify bull/bear/sideways markets?

Strategy Selection

Does it pick the right options strategy for each condition?

Position Sizing

Does it scale position size with conviction and volatility?

Exit Timing

Does it know when to take profits vs let winners run?

Avoiding Overtrading

Training Accuracy Analysis

Does it correctly output HOLD when no good setup exists?

5. How to Truly Evaluate the Model

The only reliable evaluation is paper trading performance over 2+ weeks:

Metric: Win rate, profit factor, max drawdown, Sharpe ratio

Benchmark: Must beat buy-and-hold SPY over the same period

Minimum: Must not lose more than 5% of paper trading capital

6. Summary

89.2% accuracy is a good training metric but says nothing about profitability.

The model learned the synthetic data patterns well. The real question is whether those patterns transfer to live markets. Only paper trading can answer that. Deploy now, measure real performance, then use the \$20 Modal budget to retrain on actual trading outcomes.